

Article

Design, Fabrication, and In-Flight Demonstration of a 24S NCM Battery System for an eVTOL Aircraft

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Abstract

Reliable pack-level battery systems capable of safely handling instantaneous high-C-rate discharge above 10C during take-off, climb, and hovering are required for the commercialization of urban air mobility (UAM) aircraft. However, pack-level studies on wide-range C-rate characteristics of battery systems for UAM applications remain very limited, and most previous studies have been restricted to single-cell experiments or battery-pack simulations. In this study, a 24S1P test battery pack using nickel–cobalt–manganese (NCM) pouch cells, with a nominal voltage of 88.8 V and a capacity of 22 Ah, was designed and fabricated. A two-level battery management system (BMS) based on the LTC6803G-4 was also developed. To evaluate the charge–discharge characteristics of the battery system, constant-current discharge tests were conducted under five conditions ranging from 0.2C (4.4 A) to 10.68C (235 A), and charging tests were performed over the range of 0.2C–2C. The discharge test results showed that the capacity retention remained within 97.5–100.0% in the 1C–5C range, confirming excellent power capability. Continuous discharge operation was confirmed at 10.68C, the maximum discharge condition considered for vertical take-off and climb. Under this condition, the capacity decreased to 16.26 Ah, corresponding to 74.2% of the rated capacity, owing to internal-resistance-induced voltage drop, electrochemical polarization, and early attainment of the cut-off voltage. The Peukert exponent was estimated to be 1.113. An apparent pack-level direct-current internal resistance (DCIR) of approximately 40.3 mΩ was estimated from the initial voltage-drop analysis under different discharge-current conditions. In addition, the maximum temperature during 10.68C discharge was measured as 55.1 °C, providing a thermal margin of 4.9 °C relative to the operational temperature limit of 60 °C adopted in this study. Finally, a 24S4P battery system with a capacity of 88 Ah, consisting of four 24S1P battery packs connected in parallel, was installed in the VS-210, a 210 kg-class maximum take-off weight (MTOW) eVTOL aircraft. An in-flight test was conducted by repeating six take-off–hovering–landing cycles during a total test session of 15 min 20 s, and a stable propulsion power supply was maintained throughout all flight cycles. This study provides experimental baseline data for the design and preliminary safety assessment of high-power battery systems for UAM applications by presenting both the electrical and thermal characteristics of a 24S NCM battery pack over a wide discharge-rate range of 0.2C–10.68C and in-flight eVTOL data.



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Keywords: urban air mobility; lithium-ion battery pack; high-C-rate discharge; Peukert equation; battery management system; thermal behavior; eVTOL aircraft; in-flight demonstration

1. Introduction

Urban air mobility (UAM) is an emerging air transportation system that uses electric vertical take-off and landing (eVTOL) aircraft together with advanced operation and traffic-management technologies to transport passengers and cargo within urban and sub-urban areas. UAM has attracted considerable attention as a new mobility solution that can alleviate ground traffic congestion and improve regional connectivity. Recently, active efforts have been made not only in the development of eVTOL aircraft but also in the establishment of integrated systems involving certification, vertiports, charging infrastructure, operation management, and safety validation [1,2]. As the UAM market is expected to continue expanding [3], the development of battery systems that can simultaneously provide high power density and high energy density while maintaining safety and reliability under repeated take-off, landing, and hovering conditions has become a key challenge for commercialization.

The flight mission of UAM eVTOL aircraft generally consists of vertical take-off, climb, cruise, descent, hovering, and landing phases, and the power and energy requirements differ significantly among these phases. In particular, during vertical take-off, climb, and hovering, the propulsion system must directly support the aircraft weight, requiring high power over a short period. In contrast, during cruise, the battery must supply energy for an extended duration within a limited battery mass, while during landing, the battery must still provide sufficient power even at a reduced state of charge. Therefore, UAM batteries must simultaneously achieve high specific energy, high specific power, power capability at a low state of charge, fast-charging capability, cycle life, and high safety. Yang et al. reported that, because of the unique mission profile of eVTOL aircraft, their batteries are subject to more stringent requirements than electric-vehicle batteries in terms of specific energy, specific power, fast charging, lifetime, and safety [4]. However, high specific energy and high specific power generally involve a trade-off. High-energy cells can increase stored energy by using thick electrodes and a high active-material fraction; however, under high-rate discharge, ion-diffusion limitations and concentration polarization can increase, leading to voltage drop and reduced usable capacity.

Conversely, high-power cells employ thin electrodes and low-resistance structures to enable rapid ion and electron transport, which can be disadvantageous in terms of energy storage per unit mass. Because of these characteristics, cell selection and pack design for UAM batteries cannot be based solely on rated capacity or energy density. Instead, mission-based optimization is required, considering flight range, payload, maximum power, hovering time, and operational frequency. Ayyaswamy et al. reported that the aircraft configuration and mission profile of eVTOL vehicles directly affect the electrochemical and thermal behavior inside the battery, emphasizing the need for battery designs optimized for aircraft and operating conditions [5]. The importance of cell and system design considering the trade-off between battery performance and safety in eVTOL applications has also been reported [6].

High-rate discharge during eVTOL missions can increase the initial voltage drop and Joule heating caused by battery internal resistance, thereby reducing the available discharge capacity and energy. Because Joule heating is proportional to the square of current, the high currents required during take-off and climb can cause rapid temperature rise even over a short period. In addition, high-rate operation can accelerate degradation mechanisms such as electrode-particle cracking, solid-electrolyte interphase damage, and lithium plating, and these effects can be further intensified depending on temperature and depth of discharge. Dixit et al. demonstrated, through 15C pulse-discharge tests simulating the initial climb phase of eVTOL operation, that high-power operation significantly affects cell voltage response and performance retention [7]. Recently published high-power lithium-ion cell

datasets have also reported that high C-rate, temperature, and depth of discharge are major factors accelerating cell-to-cell performance variation and degradation [8].

At the pack level, variations in cell capacity, internal resistance, and heat-transfer conditions can cause additional performance degradation and safety concerns. Even under the same current, cells with higher internal resistance exhibit greater voltage drop and heat generation, and premature over-discharge or local overheating of a specific cell can limit the usable power and energy of the entire pack. In addition, the contact resistance of busbars, connectors, and welded joints can generate additional power loss and heat under high-current conditions. Because UAM aircraft are strongly constrained not only by battery mass but also by the mass and power consumption of the cooling system, voltage drop, heat generation, and cooling performance must be considered simultaneously when sizing the battery system. Park et al. pointed out that if voltage drop, heat generation, and cooling systems are not considered together during eVTOL battery sizing, significant discrepancies can occur between conceptual design and detailed design [9]. It has also been reported that battery specific energy directly affects the range, payload, and operational feasibility of eVTOL aircraft [10].

Finally, UAM batteries are not merely energy-storage devices, but critical aviation systems directly related to propulsion and flight safety. Therefore, the battery management system must monitor cell voltage, current, and temperature in real time; estimate the state of charge and state of health; and rapidly detect overcurrent, overheating, over-discharge, and insulation abnormalities. In particular, battery performance degradation during flights can lead to propulsion-power reduction or emergency landing. Therefore, functions such as remaining flight-time prediction, power limitation, fault isolation, and thermal-runaway propagation mitigation are required [11]. However, most publicly available studies remain focused on single-cell tests, limited mission profiles, or analytical models, and studies that simultaneously present wide-range discharge characteristics, thermal behavior, BMS operation, and actual eVTOL flight data for multi-series battery packs are still limited. Therefore, for the practical implementation of UAM battery systems, electrical, thermal, and safety validation reflecting actual pack structures and flight mission conditions is required beyond cell-level performance evaluation.

In this study, a 24S NCM battery pack and a two-level battery management system (BMS) for the VS-210 eVTOL aircraft were designed and fabricated, and the system performance was evaluated through wide-range pack-level discharge tests and in-flight testing. Constant-current discharge tests from 0.2C to 10.68C were performed using a 24S1P unit battery pack to analyze the voltage, capacity, and temperature characteristics according to discharge rate. From the test results, a Peukert exponent of 1.113 and an apparent pack-level DCIR of approximately 40.3 m Ω were estimated. In addition, based on the maximum temperature of 55.1 °C measured during 10.68C discharge, a thermal margin of 4.9 °C was evaluated relative to the 60 °C temperature limit adopted in this study. Subsequently, a 24S4P system consisting of four 24S1P battery packs connected in parallel was installed in the VS-210 eVTOL aircraft. An in-flight test was conducted by repeating six take-off–hovering–landing cycles during a total test session of 15 min 20 s, and the battery voltage response and maintenance of propulsion power supply during repeated flight cycles were confirmed. This study provides an experimental basis for the design and evaluation of high-power battery systems for UAM applications by quantifying the wide-range discharge characteristics, internal resistance, and thermal behavior of a 24S NCM battery pack at the pack level and linking them with actual eVTOL in-flight results.

2. Development of the 24S NCM Battery System for an eVTOL Aircraft

2.1. VS-210 eVTOL Aircraft and Battery Requirements

The VS-210 is a single-passenger electric vertical take-off and landing (eVTOL) aircraft developed by VSPACE. It adopts a quadrotor configuration in which each of the four arms is equipped with one propulsion motor and rotor. The maximum take-off weight (MTOW) is 210 kg, while the aircraft weight and payload are 140 kg and 70 kg, respectively. The propulsion system consists of four EAGLEPOWER EA180 KV45 motors and four EP-260A-HV electronic speed controllers (ESCs), providing a maximum combined output of 96 kW under a 24S voltage condition. The battery system is installed in a dedicated bay beneath the seat using a slide-in removable structure, enabling rapid replacement and maintenance.

Because all propulsion motors of the VS-210 require high power simultaneously during vertical take-off and hovering, the battery system must provide sufficient power capability to respond to instantaneous high-rate discharge and rapid load variations. In addition, considering the limited installation space and aircraft weight constraints, the battery system must have high energy density, low internal resistance, effective thermal management, and reliable battery management functions.

The specifications of the battery system were determined based on the following design considerations. First, the number of series-connected cells was selected as 24S based on the maximum allowable voltage of the ESCs, corresponding to a full-charge voltage of 100.8 V. This is because, under the same power demand, a higher system voltage reduces the required current, thereby decreasing wiring losses and harness weight. Second, the number of parallel packs was determined based on propulsion bench-test data and measured load-current data. In the bench test of the EA180 KV45 motor with the UF6298 propeller, the thrust required per motor for MTOW hovering, 52.5 kgf, corresponded to approximately 62% throttle and a current of approximately 80 A per motor, or approximately 320 A for the overall system. To support the battery system design, a preliminary MTOW flight test was conducted using the VS-210 aircraft to measure the maximum propulsion-system load current. In this flight test, the instantaneous peak system current was approximately 550 A, which occurred during take-off, climb, and maneuvering-correction periods. Based on this maximum current, connecting four 22 Ah packs in parallel gives a current of approximately 137.5 A per pack, corresponding to approximately 6.3C. This provides a current margin of approximately 1.7 relative to the pack discharge capability evaluated in the 10.68C (235 A) discharge test in this study.

Third, the system energy was determined based on the target flight time. The nominal energy of the 24S4P configuration is 7.81 kWh (88 Ah), which is sufficient for more than 15 min of flight under partially loaded hovering conditions, corresponding to a take-off mass of 140 kg and an estimated power consumption of approximately 16–21 kW. Under MTOW normal hovering conditions, estimated to require approximately 30–39 kW based on momentum-theory scaling, the system is expected to provide approximately 10 min of flight. Based on these design considerations, a 24S4P NCM lithium-ion battery system with a nominal voltage of 88.8 V, a rated capacity of 88 Ah, and a measured mass of 43.8 kg was applied.

The main specifications and design of the VS-210 eVTOL aircraft and battery system are summarized in Table 1 and Figure 1, respectively.

Table 1. Main specifications of the VS-210 eVTOL aircraft and battery system.

Item	Specification	Remarks
Maximum take-off weight (MTOW)	210 kg	—
Aircraft weight	140 kg	Including battery system
Payload	70 kg	Single-passenger configuration
Rotor configuration	4 arms 4 rotors	Quadrotor
Propulsion motor	EA180 KV45 $\times$ 4	EAGLEPOWER
Electronic speed controller (ESC)	EP-260A-HV $\times$ 4	260 A, up to 24S
Propeller	UF6298 (62-inch) $\times$ 4	EAGLEPOWER
Maximum output power	96 kW	Total output power
Battery system	24S4P NCM Li-ion	4 of 24S1P packs in parallel
Nominal battery voltage	88.8 V	Full-charge voltage: 100.8 V
Rated battery capacity	88 Ah	22 Ah per unit pack
Nominal battery energy	7.81 kWh	88.8 V $\times$ 88 Ah
Battery system mass	43.8 kg (measured)	Approximately 20.9% of MTOW

**Figure 1.** Design of the VS-210 eVTOL aircraft manufactured by V-AEROSPACE Co., Ltd. (Incheon, Republic of Korea).

To characterize the propulsion-system performance, bench-test data for the EA180 KV45 motor combined with the UF6298 62-inch propeller are presented in Table 2. Under the 24S voltage condition of 93.6 V, the maximum thrust per motor was 98.5 kgf at 100% throttle and 245 A. Accordingly, the combined maximum thrust of the four motors was approximately 394 kgf, corresponding to a thrust-to-weight ratio (T/W) of approximately 1.88 relative to the MTOW of 210 kg. The 100% throttle condition is intended only for short-duration operation because of thermal limitations.

Table 2. Bench test performance of the EA180 KV45 motor with the UF6298 propeller (24S1P).

Throttle (%)	Thrust (kgf)	Current (A)	Speed (RPM)	Power (W)	Thrust Efficiency (gf/W)
50	34.85	43	1821	4024.8	8.66
60	49.23	72	2159	6739.2	7.31

Table 2. *Cont.*

Throttle (%)	Thrust (kgf)	Current (A)	Speed (RPM)	Power (W)	Thrust Efficiency (gf/W)
70	65.26	111	2442	10,389.6	6.28
80	82.86	162	2771	15,163.2	5.46
90	91.32	205	2820	19,188	4.76
100	98.53	245	2995	22,932	4.30

2.2. Design of the 24S NCM Battery Pack

NCM pouch-type lithium-ion cells, model P1190190S2-22Ah, supplied by Guizhou Jarwin Technology Co., Ltd., were used in this study. Only one type of cell was used to fabricate both the 24S1P test battery pack and the 24S4P onboard battery system. According to the manufacturer's specification, the cell has a nominal voltage of 3.7 V, a rated capacity of 22 Ah measured at 0.5C discharge, and a minimum capacity of 21.95 Ah. The nominal voltage of the 24S1P unit pack, consisting of 24 cells connected in series, is 88.8 V, and the full-charge and cut-off voltages are 100.8 V and 74.4 V, respectively. For the VS-210 aircraft, a 24S4P battery system was applied by connecting four 24S1P unit packs in parallel. Accordingly, the rated capacity and nominal energy of the system are 88 Ah and 7.81 kWh, respectively. In addition, a separate 24S1P test battery pack, having the same cell type and series configuration as the onboard system, was fabricated to evaluate the pack-level charge–discharge characteristics. The test pack was used exclusively for ground charge–discharge tests, whereas only the 24S4P onboard battery system was used for the in-flight test. Because the 24S1P test pack has the same cells and series configuration as the onboard system, it was used to evaluate the voltage response and C-rate dependence. However, because the housing and cooling method differs from those of the onboard system, the thermal behavior measured in the ground tests cannot be directly regarded as identical to the thermal behavior under onboard flight conditions.

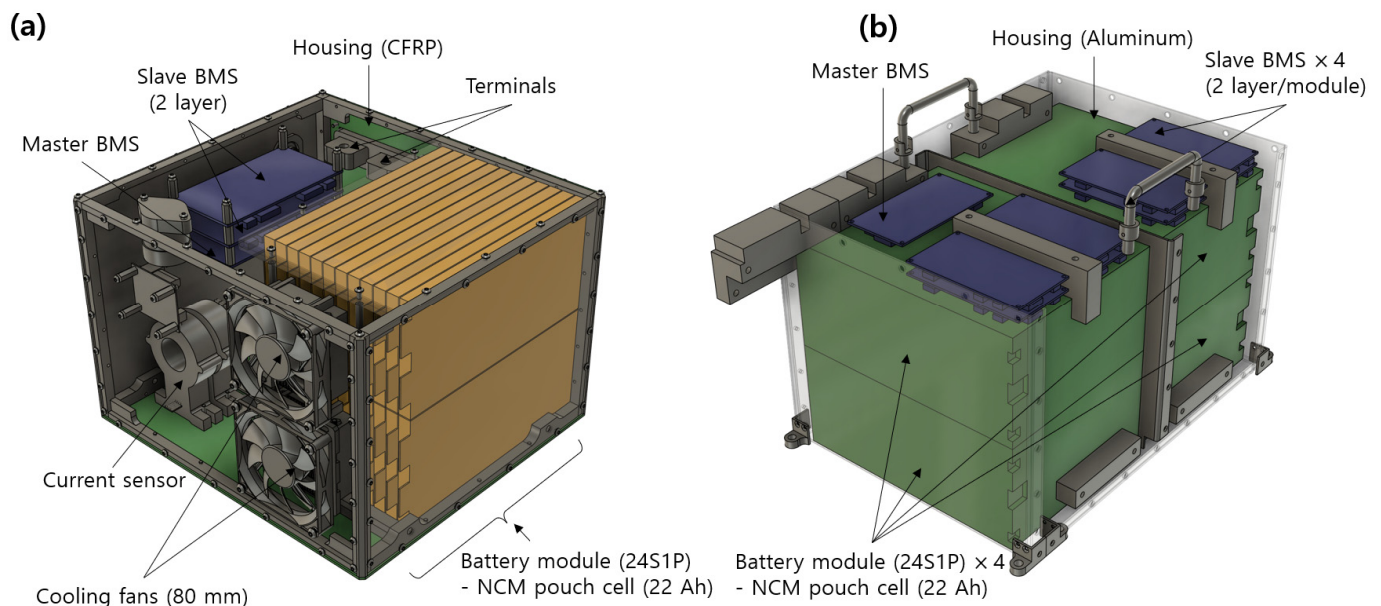
The 24S1P test battery pack was fabricated using an in-house lightweight modular structure. The NCM pouch cells were vertically stacked inside a carbon fiber reinforced plastic (CFRP) housing, and 1.5 mm-thick aluminum heat spreaders were placed between the cells to distribute heat among the cells and mitigate local temperature rise. The output terminals were configured with copper busbars to minimize contact resistance and power loss under high-current conditions.

Two 80 mm cooling fans were installed at the front of the test pack to manage the temperature by forced-air cooling. In contrast, the onboard 24S4P system was designed with an aluminum-alloy housing and a ventilated mesh structure to utilize the external airflow generated during flight for cooling. This structure allows heat inside the pack to be released to the outside without requiring a large additional cooling device.

The difference in housing and cooling methods between the test pack and the onboard system is due to differences in their operating purposes and environments. The test pack was designed for independent use during ground charge–discharge tests, handling, and transportation; therefore, a lightweight CFRP housing and integrated cooling fans were adopted to ensure stand-alone operability. In contrast, the onboard system requires fastening rigidity with the aircraft structure and can utilize rotor downwash and ram air during flight for cooling. Therefore, an aluminum housing capable of sharing structural loads and a ventilated mesh structure were adopted. The main specifications and configurations of the test pack and onboard battery system are summarized in Table 3 and Figure 2, respectively.

Table 3. Specifications of the battery pack and onboard battery system.

Item	Test Battery Pack (24S1P)	Onboard Battery System (24S4P)
Cell type	NCM pouch cell	NCM pouch cell
Nominal voltage	88.8 V	88.8 V
Full-charge voltage/cutoff voltage	100.8 V/74.4 V	100.8 V/74.4 V
Rated capacity	22 Ah	88 Ah
Nominal energy	1.95 kWh	7.81 kWh
Maximum current	235 A (10.68C)	940 A (Calculated from 4×235 A)
Continuous allowable current	110 A (5C, experimentally validated)	440 A (Calculated from 4×110 A)
Pack mass	13.5 kg	43.8 kg
Pack dimensions (L $\times$ W $\times$ H)	260 $\times$ 283 $\times$ 194 mm	390 $\times$ 290 $\times$ 260 mm
Specific energy	144.7 Wh/kg	178.4 Wh/kg
Volumetric energy density	136.9 Wh/L	265.7 Wh/L
Specific power based on continuous current	724 W/kg (110 A)	892 W/kg (440 A)
Battery mass fraction relative to MTOW	For bench testing only	20.9%
Housing material	CFRP	Aluminum Alloy and CFRP
Cell heat-spreading structure	1.5 mm aluminum heat spreader	Ventilated module structure
Cooling system	Forced air cooling with two 80 mm fans	Natural convection and flight-induced airflow

**Figure 2.** 3D modeling of the 24S NCM battery system for an eVTOL aircraft. (a) Test battery pack (24S1P); (b) onboard battery system (24S4P).

To clarify the relative position of the developed 24S NCM battery system, Table 4 compares the present system with recently reported eVTOL/UAM battery studies at the module, pack, and system levels.

Table 4. Comparison of reported eVTOL/UAM battery systems and the present 24S NCM battery system.

Study	Battery Level/Configuration	Specific Energy/Discharge Capability	Thermal Management
This work	Fabricated 24S1P test pack and 24S4P onboard system	144.7 Wh/kg for 24S1P; 178.4 Wh/kg for 24S4P; 10.68C maximum tested discharge	Forced-air cooling for test pack; natural/flight-induced airflow for onboard system
Lee [12]	High-nickel NCM pouch-cell; Fabricated 14S1P module	300 Wh/kg at cell level; 6C–8C flight scenarios	Heat sink and thermal barrier pad
Park et al. [9]	eVTOL battery-system sizing	Specific energy reduction of 4.6% when thermal constraints were considered	Air-cooled battery system
Park and Yee [13]	eVTOL battery-system sizing	125 Wh/kg at battery-system level	Liquid-cooled battery system

Compared with previous studies that mainly focused on module-level electro-thermal simulation or battery-system sizing, the present study provides a fabricated 24S NCM battery system with measured pack-level charge–discharge data and an in-flight demonstration using an actual eVTOL aircraft. However, direct comparison should be interpreted with caution because the reported specific energy and discharge conditions are provided at different levels, including cell, module, pack, and battery-system levels.

2.3. Design of the Battery Management System (BMS)

The developed battery management system adopted a two-level architecture consisting of two 12S slave boards and one master control board. Each slave board monitors 12 series-connected cells, and the two boards together measure the voltages of a total of 24 cells. The master board integrates the cell-voltage and temperature data collected from the slave boards and transmits the pack voltage, current, state of charge, and fault status to the flight controller and ground monitoring system.

The LTC6803G-4 from Analog Devices was used for the slave boards. The LTC6803G-4 can measure the voltages of up to 12 series-connected cells with a single IC and completes cell-voltage measurement within approximately 13 ms using a 12-bit ADC. A passive balancing circuit consisting of a BSS308P MOSFET and a 33 Ω resistor was applied to each cell to reduce cell-to-cell voltage deviation. The two slave boards are connected in an SPI daisy-chain configuration, allowing the master board to collect voltage data from all 24 cells through a single communication sequence.

The master board uses a GD32F405RGT6 MCU based on the ARM Cortex-M4 architecture. Two SI8652BD digital isolators were placed between the master and slave boards to electrically isolate the high-voltage cell-side circuitry from the low-voltage control-side circuitry. Communication with the flight controller was implemented using CAN 2.0B based on a TJA1042 CAN transceiver. The pack voltage, current, state of charge (SOC), maximum temperature, and protection status were configured to be transmitted at intervals of 100 ms. Major operating histories and fault information are stored in a 24LC64 EEPROM.

The protection functions of the BMS include cell overvoltage and undervoltage protection, pack overcurrent protection, overtemperature protection, and short-circuit protection. The slave boards monitor the overvoltage and undervoltage conditions of each cell, whereas the master board determines overcurrent, overtemperature, and short-circuit conditions and executes the corresponding protection actions. The threshold values for each protection function are set in the firmware and can be adjusted according to operating conditions.

In addition, when the maximum temperature exceeds 65 °C beyond the pack operating temperature limit of 60 °C adopted in this study, the BMS determines an overtemperature condition and initiates protection operation.

Figure 3 shows the real-time monitoring data acquired from the developed BMS. Through CAN communication, the BMS measured each cell voltage, pack voltage, SOC, cell temperature, and charge/discharge current in real time, and the monitoring function was confirmed to operate normally during the tests. The main hardware components and functions of the BMS are summarized in Table 5.

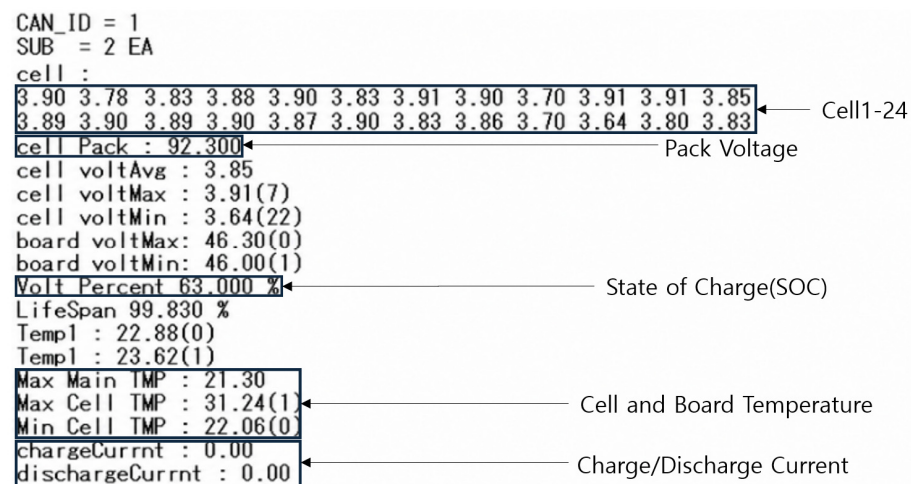


Figure 3. The sampled real-time monitoring data acquired from the developed BMS.

Table 5. Main specifications of the two-level battery management system.

Item	Slave Board (×2)	Master Board (×1)
Main IC	LTC6803G-4	GD32F405RGT6 (ARM Cortex-M4)
Processing structure	12 cells per board	Integrated processing of 24 cells
ADC	12-bit (approx. 1.5 mV resolution)	MCU-based processing
Cell voltage measuring time	Approx. 13 ms for 12 cells	Data integration
Communication	SPI daisy chain (SCKI/SDI/SDO/CSBI)	Isolated SPI/CAN 2.0B
Signal isolation	-	SI8652BD × 2, EL357N photocoupler × 3
Passive balancing	BSS308P + 33 Ω + 0.47 μH × 12	Balancing control
Temperature measurement	Two NTC channels (VTEMP1/2), LT6004 buffer	Integrated processing
Protection functions	Cell OVP/UVF	OCP/OTP/SCP
Protection devices	-	1P (330 A), 4P (1320 A)
External communication	-	TJA1042 CAN transceiver
Data storage	-	24LC64 EEPROM (64 kbit, I2C)
Power supply	Cell-stack supply (LTC6803 internal regulator)	12 V → 5 V isolated DC-DC converter, ALP1205S → 3.3 V

3. Experimental Methods

3.1. Battery Charge–Discharge Tests

To evaluate the charge–discharge performance of the battery, a 24S1P NCM lithium-ion test battery pack with a nominal voltage of 88.8 V and a rated capacity of 22 Ah was used as the test specimen. The 24S1P test pack has the same cell type and series configuration as the onboard 24S4P system; therefore, it was used to evaluate the voltage response and C-rate dependence. However, because the housing and cooling method differ from those of the onboard system, the thermal behavior measured in the ground tests cannot be directly generalized to the onboard flight conditions. The discharge tests were conducted under five constant-current conditions: 0.2C, 1C, 2C, 5C, and 10.68C. The corresponding discharge currents were 4.4 A, 22 A, 44 A, 110 A, and 235 A, respectively.

Before each discharge test, the battery pack was charged to the full-charge voltage of 100.8 V using the 0.2C constant-current–constant-voltage (CC–CV) method. During the constant-voltage stage, charging was terminated when the charging current decreased below 0.05C, after which the pack was rested for 30 min to stabilize the voltage and temperature. The discharge tests were performed under the specified constant-current conditions and were automatically terminated when the pack voltage reached the cut-off voltage of 74.4 V. Charging tests were also conducted using the same battery pack at 0.2C, 1C, and 2C, corresponding to charging currents of 4.4 A, 22 A, and 44 A, respectively. Charging was performed using the CC–CV method. When the pack voltage reached 100.8 V during the constant-current stage, the charging mode was switched to the constant-voltage stage, and the test was terminated when the charging current decreased below 0.05C.

All charge–discharge tests were conducted indoors at an ambient temperature of 15–23 °C. During the tests, two 80 mm cooling fans installed at the front of the unit battery pack were operated under their rated conditions. The pack voltage, current, and temperature were recorded using the BMS at a sampling rate of 1 Hz. The cell temperature was measured using an NTC sensor connected to the BMS. The sensor was attached to the surface of a representative cell located near the center of the 24S1P test battery pack. Therefore, the measured temperature represents the surface bulk temperature of the central cell, rather than the average temperature of the entire pack or the temperature of each individual cell. The charge–discharge time, capacity, energy, voltage drop, and maximum temperature were evaluated for each test. A schematic diagram of the charge–discharge test configuration for the test battery pack is shown in Figure 4.

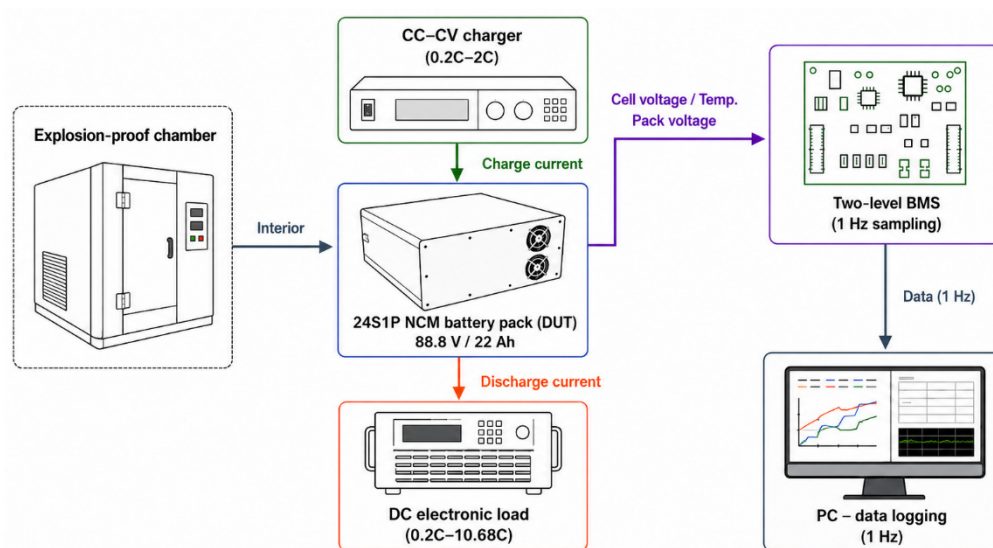


Figure 4. Schematic diagram of the battery pack charge–discharge test configuration.

The discharge capacity Q_d and discharge energy E_d were calculated as follows.

$$Q_d = \frac{1}{3600} \int_0^{t_f} I(t) dt \quad (1)$$

$$E_d = \frac{1}{3600} \int_0^{t_f} V(t) I(t) dt \quad (2)$$

where $I(t)$ is the discharge current (A), $V(t)$ is the pack voltage (V), and t_f is the discharge end time (s). The factor $1/3600$ in Equations (1) and (2) is used to convert the time unit from seconds to hours.

The ohmic heat generated by the internal resistance of the pack, Q_{ohm} , was estimated as follows.

$$Q_{ohm} = \int_0^{t_f} I^2(t) R_i dt \quad (3)$$

Assuming that the internal resistance remains constant during constant-current discharge, Equation (3) can be simplified as follows.

$$Q_{ohm} = I^2 R_i t_f \quad (4)$$

The decrease in effective capacity with increasing discharge rate was analyzed using the Peukert relationship [14].

$$t = K I^{-k} \quad (5)$$

This relationship can be expressed in logarithmic form as follows.

$$\ln(t) = \ln(K) - k \ln(I) \quad (6)$$

where k is the Peukert exponent, and K is an empirical constant. The Peukert exponent was determined by linear regression of $\ln(I)$ and $\ln(t)$.

The apparent pack-level direct-current internal resistance (DCIR) was estimated using the initial voltage drop and current difference measured under different discharge-current conditions, as follows [15].

$$R_i \approx \frac{\Delta V}{\Delta I} \quad (7)$$

The initial state of charge, rest time, and voltage measurement point were kept as consistent as possible during the DCIR estimation.

3.2. eVTOL In-Flight Test

To evaluate the applicability and operational feasibility of the developed battery system under actual flight conditions, an in-flight test was conducted using the VS-210 eVTOL aircraft equipped with the 24S4P NCM lithium-ion battery system. Figure 5 shows photographs of the VS-210 eVTOL aircraft and onboard battery system (24S4P). The test aircraft was equipped with a 24S4P NCM lithium-ion battery system composed of four 22 Ah packs connected in parallel as shown in Figure 5b. The nominal voltage and rated capacity of the battery system were 88.8 V and 88 Ah, respectively, and the nominal energy was approximately 7.81 kWh. At the time of the test, the ambient temperature was approximately 5 °C, and the wind speed was approximately 2–3 m/s. The flight profile was set as static hovering after take-off, with the aircraft maintaining a constant altitude, and the battery voltage variation was measured during take-off, hovering, and landing.



Figure 5. Photographs of the VS-210 eVTOL aircraft and battery system. (a) VS-210 eVTOL aircraft; (b) onboard battery system (24S4P).

The voltage of each unit battery pack was simultaneously recorded using a data logger at a sampling rate of 10 Hz. From the measured data, the instantaneous voltage drops immediately after take-off, the voltage-decrease characteristics during hovering, the total hovering flight time, the voltage response of the battery system during actual flight, and the maintenance of propulsion power supply were evaluated.

The in-flight test was conducted for a total duration of 15 min 20 s. Based on the measured voltage data, the voltage drop during the take-off and landing phases and the voltage-change-rate characteristics during steady hovering were analyzed.

4. Results and Discussion

4.1. Discharge Voltage Characteristics

Figure 6 shows the voltage–time curves of the 24S1P NCM battery pack at different discharge C-rates. The tests were conducted under five constant-current conditions: 0.2C, 1C, 2C, 5C, and 10.68C. Under all discharge conditions, a common voltage behavior was observed, consisting of an initial voltage drop immediately after the start of discharge, a relatively gradual middle-voltage region, and a rapid voltage decrease near the end of discharge [16]. As the discharge rate increased, the initial voltage drop became larger and the middle-voltage region became shorter. This tendency is attributed to increased ohmic loss due to internal resistance and electrochemical polarization. This three-stage voltage behavior is consistent with the typical high-rate discharge response of lithium-ion batteries reported in the literature. Under ultra-high discharge-rate conditions, voltage deterioration is affected not only by ohmic resistance but also by charge-transfer polarization, ionic transport resistance, solid-state lithium diffusion limitation, and temperature-dependent electrochemical kinetics [17]. These effects become more pronounced at 10.68C, leading to a larger voltage drop, a shortened middle-voltage region, and earlier cut-off. Such voltage deterioration can be mitigated by cell- and system-level improvements, including electrode-structure optimization, reduced tab/busbar/contact resistance, improved current distribution, and enhanced thermal management [18]. However, high-power-oriented cell designs should be balanced with specific energy requirements for eVTOL applications.

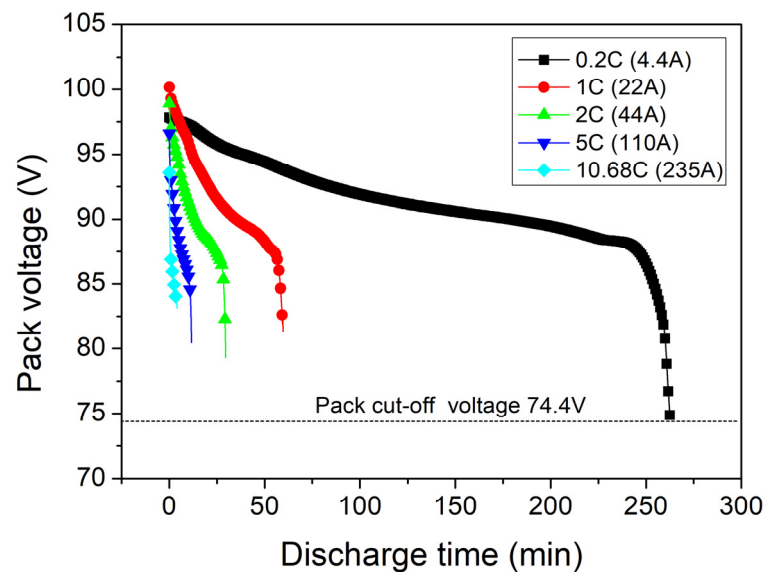


Figure 6. Pack voltage characteristics of the 24S1P NCM lithium-ion battery pack at different discharge C-rates.

Under the 1C condition, discharge started from an initial voltage of 100.16 V and ended at 81.32 V after 59 min 45 s. Using the initial voltage difference of $\Delta V = 3.55$ V and the current difference of $\Delta I = 88$ A between the 1C and 5C tests, the apparent direct-current internal resistance (apparent DCIR) of the pack was estimated from Equation (7) as follows:

$$R_i \approx \frac{\Delta V}{\Delta I} = \frac{3.55}{88} \approx 40.3 \text{ m}\Omega \quad (8)$$

This value does not represent a standard DCIR measured by applying a current pulse at the same SOC. Instead, it represents a pack-level equivalent internal resistance estimated using the initial voltage difference and current difference measured under different discharge conditions. Therefore, the estimated value includes not only the internal resistance of the 24 cells but also the contact resistances of the busbars, connection parts, and connectors.

Under the 10.68C condition, discharge started at an initial voltage of 93.59 V with a current of 235 A and ended at 83.15 V after 4 min 9 s. At the midpoint of discharge, approximately 2 min 4 s, the pack voltage was measured as 85.46 V, which is 3.34 V lower than the nominal voltage of 88.8 V. These results indicate that, under high-rate discharge, the initial voltage drop and polarization increase significantly, thereby limiting the usable voltage range and discharge duration.

4.2. Discharge Temperature and Ohmic Heat Generation Characteristics

Figure 7 and Table 6 show the temperature variation in the battery pack according to the discharge rate. The total heat generation of a lithium-ion battery consists of irreversible ohmic heat, electrochemical reaction heat, and reversible entropic heat [19]. In this study, the ohmic heat, which is dominant during constant-current high-rate discharge, was estimated using Equation (4). The estimated ohmic heat increased from 12.3 kJ at 0.2C to 70.0 kJ at 1C, 138.6 kJ at 2C, 341.2 kJ at 5C, and 554.7 kJ at 10.68C. Although the discharge time decreased significantly under high-rate conditions, the total heat generation continuously increased with increasing discharge rate owing to the effect of ohmic heat, which is proportional to the square of the current.

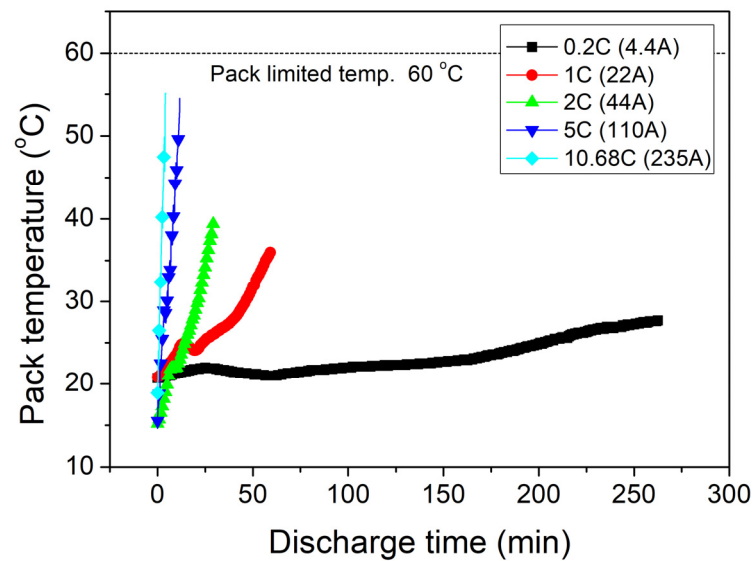


Figure 7. Pack temperature characteristics of the 24S1P NCM lithium-ion battery pack at different discharge C-rates.

Table 6. Discharge voltage and temperature results at different C-rates (24S1P).

C-Rate	Discharge Current (A)	Discharge Time	Initial Voltage (V)	Final Voltage (V)	Initial Temperature (°C)	Maximum Temperature (°C)	Temperature Rise (°C)
0.2C	4.4	4 h 22 min 33 s	97.81	74.81	20.7	27.8	7.1
1C	22	0 h 59 min 45 s	100.16	81.32	20.8	36.1	15.3
2C	44	0 h 29 min 34 s	98.91	79.31	15.2	39.8	24.6
5C	110	0 h 11 min 39 s	96.61	80.45	15.5	54.5	39.0
10.68C	235	0 h 04 min 09 s	93.59	83.15	18.9	55.1	36.2

The maximum temperature of the battery pack was measured using an NTC temperature sensor attached to the cell surface at the center of the module. The maximum temperatures were 27.8 °C, 36.1 °C, 39.8 °C, 54.5 °C, and 55.1 °C under the 0.2C, 1C, 2C, 5C, and 10.68C conditions, respectively. The temperature-rise rate was approximately 0.03 °C/min at 0.2C, whereas it increased significantly to approximately 8.72 °C/min at 10.68C. This result indicates that rapid temperature rise can occur even over a short period under high-rate discharge conditions, providing important data for establishing BMS-based temperature-dependent power-limiting and cooling-control criteria.

At 10.68C discharge, the maximum temperature was measured as 55.1 °C. Compared with the operational temperature limit of 60 °C for the battery system adopted in this study, the thermal margin was 4.9 °C. However, the local temperature near the cell tabs, welded joints, and busbar connections were not separately measured in this study. Therefore, the reported temperature does not include possible local hot spots near the external current-collecting regions. Meanwhile, as shown in Table 6, the tests were conducted at 15–23 °C and did not include summertime flight conditions, where the ambient temperature may increase to approximately 30 °C. Under such conditions, the battery internal resistance

may slightly decrease, but the thermal margin would also be reduced. Considering the approximately 36 °C temperature rise during 10.68C discharge, high-rate operation under summertime conditions could cause the cell surface temperature to approach or exceed the 60 °C operational limit, requiring BMS-based power limitation and improved cooling.

4.3. Discharge Capacity, Energy, and Peukert Analysis

Table 7 and Figure 8 show the variations in discharge capacity, discharge energy, and capacity retention ($= \frac{Q_d(C\text{-rate})}{Q_d(1C)} \times 100$) according to the discharge rate. Based on the discharge capacity of 21.91 Ah measured under the 1C condition, the discharge capacities under the 2C and 5C conditions were 21.65 Ah and 21.36 Ah, respectively, corresponding to capacity retentions of 98.8% and 97.5%. Therefore, relatively stable capacity characteristics were maintained in the 1C–5C range despite the increase in discharge rate.

Table 7. Discharge capacity, energy, capacity retention, and estimated ohmic heat at different C-rates (24S1P).

C-Rate	Current (A)	Discharge Capacity (Ah)	Discharge Energy (Wh)	Capacity Retention (%)	Estimated Ohmic Heat (kJ)
1C	22	21.91	2006.5	100.0	70.0
2C	44	21.65	1955.5	98.8	138.6
5C	110	21.36	1886.1	97.5	341.2
10.68C	235	16.26	1394.3	74.2	554.7

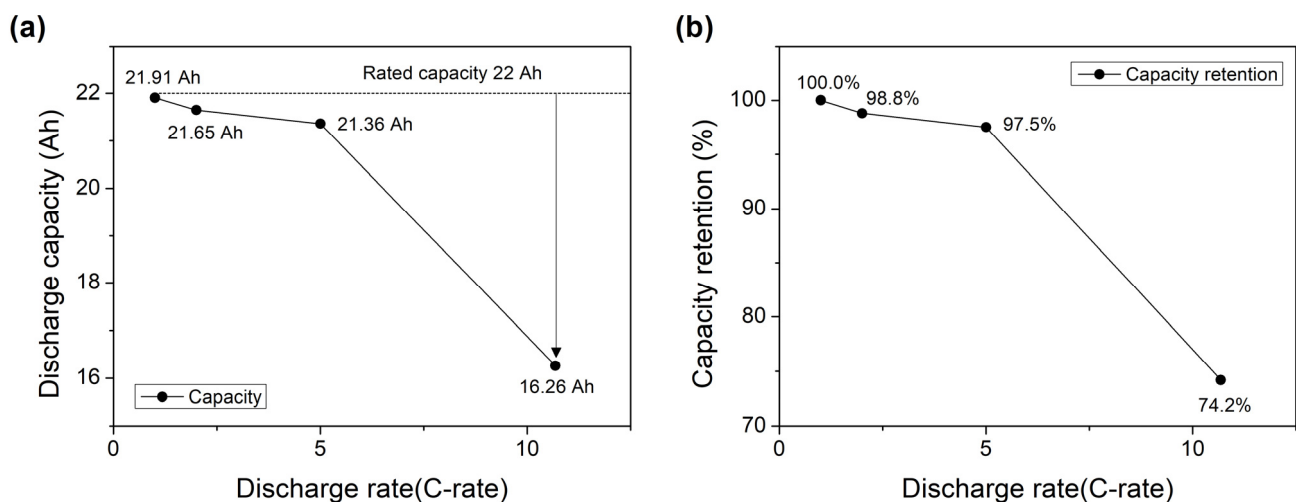


Figure 8. Discharge capacity and capacity retention at different C-rates for test battery pack (24S1P). (a) Discharge capacity; (b) capacity retention.

In contrast, under the 10.68C condition, the discharge capacity decreased to 16.26 Ah, corresponding to 74.2% of the capacity measured at 1C. This decrease can be attributed to the increased voltage drop caused by internal resistance and electrochemical polarization under high-rate discharge, which causes the cell voltage to reach the cut-off condition earlier. In addition, the effective capacity and energy are considered to decrease because lithium-ion diffusion inside the electrodes and interfacial reactions cannot sufficiently follow the high current demand [20].

By applying the 1C–10.68C discharge data to the Peukert-type empirical relationships in Equations (5) and (6), the Peukert exponent and empirical constant obtained from linear regression were determined as follows:

$$k = 1.113, K = 3843 \text{ s} \cdot \text{A}^k, R^2 = 0.9949 \quad (9)$$

The coefficient of determination, $R^2 = 0.9949$, indicates that the relationship between discharge current and discharge time within the tested range is well described by the Peukert-type empirical equation. However, because the Peukert equation was originally proposed for lead–acid batteries [14], the value of k obtained in this study should be interpreted as an empirical indicator dependent on the test temperature, cut-off voltage, and C-rate range, rather than as an intrinsic electrochemical constant of the lithium-ion battery pack.

Under the 0.2C condition, the discharge capacity was measured as 19.30 Ah, which was lower than that under the 1C condition. Considering that a higher capacity is generally expected under low-rate discharge, this result is difficult to explain simply by self-discharge or the Peukert effect. Therefore, the effects of the initial state of charge, test starting voltage, cut-off condition, measurement error during the long-duration test, and cell-balancing condition need to be further examined. Accordingly, in this study, the 0.2C result was regarded as a preliminary result requiring further verification of data reliability and was excluded from the capacity-retention and Peukert regression analyses.

4.4. Charging Characteristics

Table 8 and Figure 9 show the voltage and temperature variations in the 24S1P NCM battery pack according to the charging rate. The charging tests were conducted using the CC–CV method under 0.2C, 1C, and 2C conditions. Under all conditions, the final pack voltage was measured in the range of 100.76–100.78 V, which is close to the set full-charge voltage of 100.8 V.

Table 8. Charging results at different C-rates for test battery pack (24S1P).

C-Rate	Current (A)	Charge Time	Initial Voltage (V)	Final Voltage (V)	Max. Temperature (°C)	Temperature Rise (°C)
0.2C	4.4	5 h 10 min	78.15	100.76	22.1	0.0
1C	22	1 h 16 min	83.56	100.76	20.8	4.1
2C	44	37 min 47 s	90.26	100.78	28.7	4.8

The charging time was 5 h 10 min at 0.2C, 1 h 16 min at 1C, and 37 min 47 s at 2C, indicating that the charging time was significantly reduced as the charging rate increased. However, because the initial voltages of the tests differed from 78.15 to 90.26 V, the difference in charging time is considered to have been affected not only by the charging rate but also by the difference in the state of charge at the start of each test. Therefore, repeated tests under the same initial SOC or initial voltage condition are required to quantitatively compare the charging time according to the charging rate.

During 2C charging, the maximum temperature was 28.7 °C, and the temperature rise from the initial temperature was measured as 4.8 °C. This value is significantly lower than the temperature rise of 24.6 °C observed during the 2C discharge test. This difference can be attributed to the gradual decrease in charging current during the constant-voltage stage as the pack voltage increases during the CC–CV charging process, as well as the different contributions of irreversible heat generation and reversible entropic heat during charging

and discharging [19]. Under the present test conditions, excessive temperature rise did not occur during 2C charging, confirming the applicability of short-duration charging.

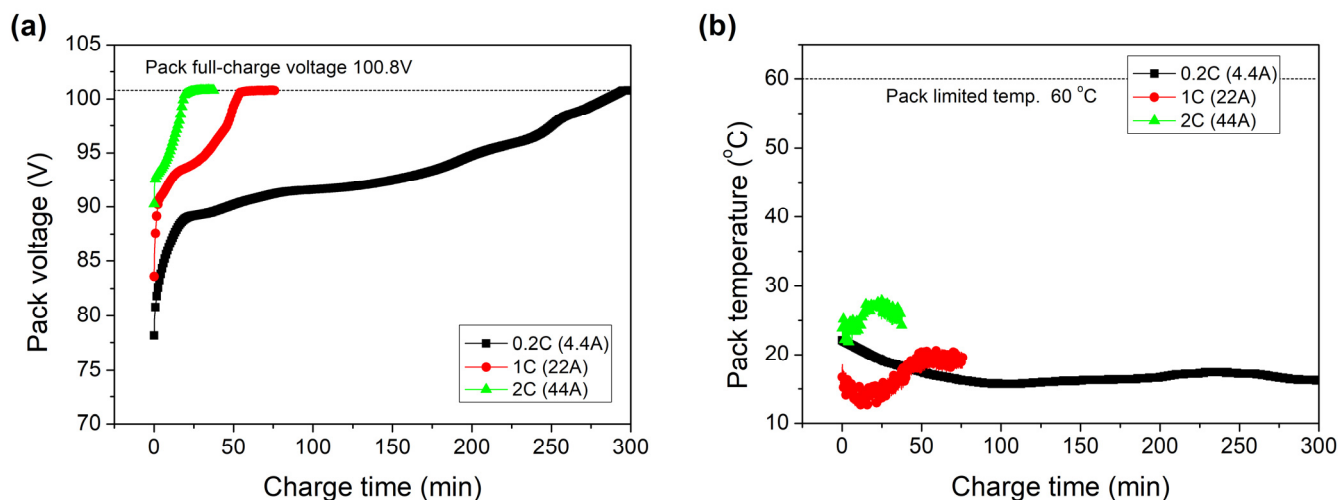


Figure 9. Charging characteristics of the 24S1P NCM lithium-ion test battery pack at different C-rates: (a) pack voltage and (b) pack temperature.

4.5. eVTOL In-Flight Performance and Discussion

Figure 10 shows the variation in battery-system voltage measured during the in-flight test session, which lasted for a total of 15 min 20 s. In this session, six flight cycles consisting of take-off, hovering, and landing were repeatedly performed, and each cycle can be clearly distinguished from the voltage profile. During the take-off and hovering, or loaded, period of each cycle, the voltage dropped sharply and remained at a lower level with fluctuations. In contrast, during the waiting, or no-load, period after landing, the voltage gradually recovered owing to polarization relaxation.

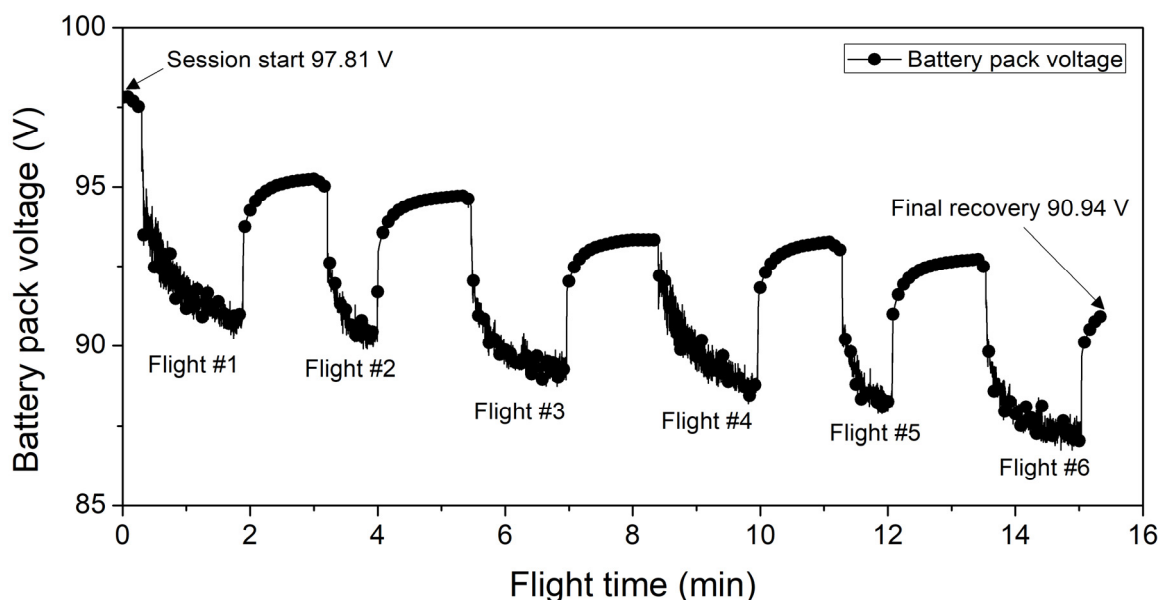


Figure 10. In-flight voltage profile of the 24S4P onboard battery system during six repeated takeoff–hover–landing cycles (15 min 20 s session, 10 Hz).

The loaded durations of the six cycles ranged from 52 to 100 s, corresponding to a cumulative loaded flight time of approximately 8.2 min. The initial voltage before the session was 97.81 V, the minimum voltage during the session was 86.75 V, observed in

the sixth cycle, and the recovery voltage after the final landing was 90.94 V. The gradual decrease in the recovery voltage after landing, from 95.27 V to 92.72 V as the cycles progressed, reflects the decrease in SOC caused by discharge.

The voltage drop of approximately 8 V at the initial take-off of the first cycle is considered to result from the combined effects of resistive voltage drop due to the rapid power demand of the propulsion motors and electrochemical polarization. However, because the voltage drop persisted for approximately 1.5 min, it is difficult to regard it as a purely instantaneous DCIR-related voltage drop. In addition, because the actual flight current was not directly measured, it is not appropriate to directly calculate the in-flight current or system DCIR using the maximum combined output of the propulsion system, 96 kW. To quantitatively evaluate the DCIR and power consumption under actual flight conditions, the pack voltage and current should be measured simultaneously at the same sampling rate.

Because the current was not directly measured during the in-flight test, the power consumption was cross estimated using two independent methods. First, a voltage-based estimation was performed. When the final recovery voltage after the session, 90.94 V, was matched with the voltage–capacity relationship of the ground 1C discharge curve, the capacity consumed during the session was estimated to be approximately 35–45 Ah. However, because the ambient temperature during flight was relatively low, approximately 5 °C, this method may overestimate the consumed capacity owing to the increase in internal resistance and the downward shift in the discharge curve.

Second, the power consumption was estimated based on the propulsion-system bench-test data. For a take-off mass of 140 kg, the required thrust per motor is 35 kgf. In the bench data shown in Table 2, this corresponds to approximately 50% throttle, approximately 43 A per motor, approximately 172 A for the overall system, and approximately 16.1 kW. Under actual flight conditions, the power consumption is expected to increase compared with the bench-test condition because of the transient currents during the six take-offs, wind correction at 2–3 m/s, and low-temperature effects.

Combining the two estimation methods, the average current during the flight, or loaded, periods was estimated to be approximately 180–240 A, corresponding to approximately 2.0–2.7C per pack. The average power consumption was estimated to be approximately 16–21 kW. Based on the cumulative loaded time of approximately 8.2 min, the consumed capacity during the session was calculated to be approximately 25–33 Ah. This corresponds to the stable discharge region below 5C confirmed in the ground tests. However, further verification using synchronized current measurements is required.

For reference, in a separate flight test conducted under the MTOW condition of 210 kg, the measured instantaneous peak system current of 550 A corresponds to approximately 137.5 A per pack in the 24S4P battery system. From the propulsion-system viewpoint, assuming equal load distribution among the four motors, it is also comparable to approximately 137.5 A per motor. According to momentum theory, in which the hovering power is proportional to $W^{1.5}$ [21], the steady hovering power under the MTOW condition is estimated to be approximately 30–39 kW, which is approximately 1.84 times higher than that in the present flight test at 140 kg, or approximately 16–21 kW. This corresponds to a system current of approximately 340–440 A. The measured peak current of 550 A is approximately 1.3–1.6 times higher than this estimated steady hovering current, which is consistent with the transient current characteristics of multirotor aircraft during take-off and climb.

Based on the bench-test data in Table 2, a system current of 550 A corresponds to 137.5 A per motor, approximately 75% throttle, and approximately 74 kgf thrust per motor. This provides an instantaneous thrust-to-weight ratio of approximately 1.4. Even under this

condition, the discharge rate per pack, approximately 6.3C, has a margin of approximately 1.7 relative to the maximum tested current confirmed in the 10.68C discharge test. The flight test video showing the repeated take-off–hovering–landing cycles of the VS-210 eVTOL aircraft is provided as Supplementary Video S1.

In the 24S4P parallel configuration, the four packs share the same output bus; therefore, the terminal voltages of the packs are inherently identical, and differences in current sharing among individual packs cannot be distinguished from voltage measurements alone. Accordingly, synchronized measurement of the current of each pack is required in future work to evaluate pack-to-pack balance. The results of the VS-210 flight test are summarized in Table 9.

Table 9. Summary of the VS-210 eVTOL flight test results.

Item	Result
Test environment	Ambient temperature of 5 °C and wind speed of 2–3 m/s
Flight profile	Six repeated take-off–hovering–landing cycles
Take-off mass	Approx. 140 kg, including 115 kg of aircraft mass and 25 kg of payload, 67% of MTOW
Total test time	15 min 20 s total session duration; 8.2 min cumulative loaded flight time
Battery configuration	24S4P (88.8 V/88 Ah)
Measurement device	Graphtec data logger
Sampling rate	10 Hz; 9208 samples during flight period
Initial voltage	97.81 V
Minimum section voltage	86.75 V, observed during the sixth cycle
Final recovery voltage	90.94 V, measured after the final landing
Voltage drops during the first lift-off	Approx. 8 V
Inter-pack voltage behavior	Identical voltage level through the shared parallel bus
Estimated average power consumption	Approx. 16–21 kW
Estimated average discharge rate	Approx. 2.0–2.7C, based on the 4P battery configuration

In contrast, the present study is distinguished from previous studies in that wide-range constant-current discharge tests from 0.2C to 10.68C were conducted using a 24S NCM battery pack, and the voltage, capacity, energy, and temperature characteristics were quantitatively evaluated at the pack level. In particular, the maximum temperature under the condition of 10.68C was measured as 55.1 °C, confirming that the thermal margin can decrease rapidly during high-rate discharge. This result indicates that not only cell performance but also busbars, connectors, housing, cooling structure, and BMS power-limiting logic should be considered together in the design of eVTOL battery systems.

In the present in-flight test, the developed 24S NCM battery system was confirmed to continuously supply propulsion power throughout the take-off, hovering, and landing phases of the VS-210 aircraft. This is meaningful in that the high-rate discharge and thermal characteristics evaluated through ground charge–discharge tests were linked with actual aircraft operation. However, the present flight test was conducted at approximately 5 °C and under mild wind conditions of 2–3 m/s; therefore, it should be regarded as an initial in-flight demonstration rather than a full validation under representative UAM operating conditions. Under low-temperature conditions, increased internal resistance and polarization may lead to larger voltage drop, reduced usable capacity and energy, and

earlier cut-off. Under high-temperature conditions, the internal resistance may slightly decrease, but the thermal margin would be reduced, requiring BMS-based power limitation and improved cooling during high-C-rate operation. In addition, strong wind or gust conditions would increase propulsion power demand for attitude stabilization, resulting in higher battery current, heat generation, energy consumption, and SOC depletion rate. Moreover, because pack current, cell temperature, and energy-consumption data were not acquired during the in-flight test, there are limitations in quantitatively analyzing power sharing and energy efficiency. In future work, the current of each pack and the voltage and temperature of each cell should be measured synchronously to more precisely evaluate power consumption, thermal behavior, and battery degradation characteristics under actual flight conditions.

5. Conclusions

In this study, a 24S NCM lithium-ion battery system for application to the VS-210 eVTOL aircraft with an MTOW of 210 kg was designed and fabricated, and its electrical and thermal performance was evaluated through ground charge–discharge tests and in-flight testing. The main findings are summarized as follows:

- (1) The battery management system was developed with a two-level architecture consisting of two 12S slave boards based on the LTC6803G-4 and a master board based on the GD32F405RGT6 MCU. SI8652BD digital isolators were applied between the master and slave boards to electrically isolate the high-voltage cell side from the low-voltage control side. In addition, the battery status information was linked with the flight controller in real time through CAN 2.0B communication.
- (2) From the constant-current discharge tests of the 24S1P test battery pack, the capacity retention based on the 1C condition was 97.5–100.0% in the 1C–5C range, indicating relatively stable discharge performance. In contrast, under the 10.68C condition, the discharge capacity decreased to 16.26 Ah, corresponding to a capacity retention of 74.2% relative to the 1C condition. When the 1C–10.68C data were applied to the Peukert-type empirical equation, the Peukert exponent was estimated as $k = 1.113$, with a coefficient of determination of $R^2 = 0.9949$. In addition, the apparent DCIR of the pack was estimated to be approximately 40.3 m Ω from the initial voltage difference under different discharge-current conditions.
- (3) The temperature rise in the battery pack increased significantly as the discharge rate increased. During 10.68C discharge, the maximum temperature was 55.1 °C, and the average temperature-rise rate was approximately 8.72 °C/min. This result indicates that the thermal margin can decrease rapidly even during short-duration high-rate discharge and suggests that securing sufficient cooling time between flights and implementing BMS-based temperature and power-limiting strategies are necessary for repeated take-off and landing operation.
- (4) In the CC–CV charging tests, the charging time and temperature rise under the 2C condition were 37 min 47 s and 4.8 °C, respectively. This temperature rise was lower than the 24.6 °C observed under the same 2C discharge condition, and excessive temperature rise did not occur during 2C charging under the present test conditions. However, because the initial voltage and initial SOC differed among the charging tests, repeated tests under identical initial conditions are required to quantitatively compare the charging time and thermal characteristics according to the charging rate.
- (5) The developed battery system was installed in the VS-210 eVTOL aircraft, and an in-flight test was conducted by repeating six take-off–hovering–landing cycles during a total test session of 15 min 20 s. The initial voltage before the session, minimum voltage during the session, and final recovery voltage were 97.81 V, 86.75 V, and

90.94 V, respectively, indicating that propulsion power supply was stably maintained throughout all cycles. A voltage drop of approximately 8 V was observed during take-off in the first cycle. The repeated execution of six consecutive take-off and landing cycles within a single session is meaningful in that it simulates multi-sortie operation of UAM aircraft. These results confirm the applicability of the developed battery system to an actual aircraft. However, because current, energy consumption, and cell temperature were not measured simultaneously during the in-flight test, quantitative analysis of power sharing and energy efficiency remains limited.

- (6) In the flight test under the MTOW condition of 210 kg, an instantaneous peak system current of approximately 550 A, corresponding to approximately 6.3C per pack, was measured. This confirms that the discharge rate under the actual maximum-load condition was within the range verified by the ground discharge test, namely 10.68C. Based on momentum theory, the steady hovering power under the MTOW condition was estimated to be approximately 30–39 kW, corresponding to 340–440 A. The measured mass of the onboard battery system was 43.8 kg, corresponding to approximately 20.9% of the MTOW, and its specific energy was approximately 178.4 Wh/kg.

This study is meaningful in that it links wide-range discharge tests from 0.2C to 10.68C with in-flight testing using a 24S NCM pouch-type battery pack and presents pack-level voltage, capacity, energy, and thermal characteristics. In particular, the capacity decrease and rapid temperature rise under high-rate discharge conditions show that not only cell performance but also busbars, connectors, housing, cooling structure, and BMS protection logic should be considered in an integrated manner when designing eVTOL battery systems.

The limitations of this study are that the tests were conducted under limited indoor-temperature conditions, and cycle degradation due to repeated charge–discharge operation and complex mission profiles, including take-off, climb, cruise, and landing, were not considered. In future work, charge–discharge tests under various temperature and SOC conditions, repeated high-rate cycle tests, and synchronized in-flight measurements of current and temperature will be conducted. In addition, a 2RC equivalent circuit model will be established, and an EKF-based SOC estimation algorithm will be applied to verify the state-estimation accuracy and BMS control performance under actual flight conditions.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/batteries12090317/s1>: Video S1: In-flight test of the VS-210 eVTOL aircraft equipped with the developed 24S4P NCM lithium-ion battery system.

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Conflicts of Interest: SuHo Yu and Yu-Jin Jung are employees of V-AEROSPACE Co., Ltd., and Bum-Dong Cho is the Chief Executive Officer of V-AEROSPACE Co., Ltd., which developed the VS-210 eVTOL aircraft used in this study. The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

ADC	Analog-to-Digital Converter
BMS	Battery Management System
CAN	Controller Area Network
CC–CV	Constant-Current–Constant-Voltage
CFRP	Carbon Fiber Reinforced Plastic
DCIR	Direct-Current Internal Resistance
DUT	Device Under Test
EEPROM	Electrically Erasable Programmable Read-Only Memory
EKF	Extended Kalman Filter
ESC	Electronic Speed Controller
eVTOL	Electric Vertical Take-Off and Landing
IC	Integrated Circuit
I2C	Inter-Integrated Circuit
MCU	Microcontroller Unit
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
MTOW	Maximum Take-Off Weight
NCM	Nickel–Cobalt–Manganese
NTC	Negative Temperature Coefficient
OCP	Overcurrent Protection
OTP	Overtemperature Protection
OVP	Overvoltage Protection
RPM	Revolutions Per Minute
SCP	Short-Circuit Protection
SOC	State of Charge
SOH	State of Health
SPI	Serial Peripheral Interface
T/W	Thrust-to-Weight Ratio
UAM	Urban Air Mobility
UVP	Undervoltage Protection

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